

New York City Schools Perception

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Research Questions

1. Is there a relationship between gender percentages and average SAT scores? How about for the different sections of the SAT (Reading, Writing, and Math)?
2. Which NYC schools seem to have the best quality metrics according to survey data? Is there a difference if you break this down by response type?
3. Can you learn anything from the differences in school quality metric perception between groups? For example, if you created a new variable by subtracting saf_p_11 from saf_s_11, do you think this difference in how students and parents perceive safety may be related to any other variables?

```
library(tidyverse)
library(janitor)
library(readr)
library(corrplot)
```

```
url <- "https://dsserver-prod-resources-1.s3.us-east-1.amazonaws.com/327/combined.csv"
df_combined <- read_csv(url)
```

```
## Rows: 479 Columns: 30
## -- Column specification -----
## Delimiter: ","
## chr (3): DBN, school_name, boro
## dbl (27): Num of SAT Test Takers, SAT Critical Reading Avg. Score, SAT Math ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
df_d75 <- read_tsv("masterfile11_d75_final.txt")
```

```
## Rows: 56 Columns: 1773
## -- Column specification -----
## Delimiter: "\t"
## chr (5): dbn, bn, schoolname, studentssurveyed, schooltype
## dbl (1739): d75, highschool, rr_s, rr_t, rr_p, N_s, N_t, N_p, nr_s, nr_t, nr...
## lgl (29): p_q5, p_q9, p_q13a, p_q13b, p_q13c, p_q13d, p_q14a, p_q14b, p_q1...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
names(df_d75)
```

```
## [1] "dbn"          "bn"           "schoolname"
## [4] "d75"          "studentssurveyed" "highschool"
## [7] "schooltype"   "rr_s"         "rr_t"
```

##	[10]	"rr_p"	"N_s"	"N_t"
##	[13]	"N_p"	"nr_s"	"nr_t"
##	[16]	"nr_p"	"saf_p_11"	"com_p_11"
##	[19]	"eng_p_11"	"aca_p_11"	"saf_t_11"
##	[22]	"com_t_11"	"eng_t_11"	"aca_t_11"
##	[25]	"saf_s_11"	"com_s_11"	"eng_s_11"
##	[28]	"aca_s_11"	"saf_tot_11"	"com_tot_11"
##	[31]	"eng_tot_11"	"aca_tot_11"	"p_q1c"
##	[34]	"p_q10a"	"p_q10b"	"p_q10c"
##	[37]	"p_q10d"	"p_q10e"	"p_q10f"
##	[40]	"p_q11a"	"p_q11b"	"p_q11c"
##	[43]	"p_q11d"	"p_q11e"	"p_q1b"
##	[46]	"p_q1e"	"p_q1f"	"p_q2a"
##	[49]	"p_q2b"	"p_q3c"	"p_q3d"
##	[52]	"p_q4a"	"p_q6c"	"p_q12c"
##	[55]	"p_q1a"	"p_q1d"	"p_q3a"
##	[58]	"p_q3b"	"p_q3e"	"p_q4b"
##	[61]	"p_q4c"	"p_q6e"	"p_q7"
##	[64]	"p_q8a"	"p_q8b"	"p_q12d"
##	[67]	"p_q1g"	"p_q6a"	"p_q6b"
##	[70]	"p_q6d"	"p_q6f"	"p_q6g"
##	[73]	"p_q6h"	"p_q12a"	"p_q12b"
##	[76]	"p_q12e"	"p_q12f"	"p_q5"
##	[79]	"p_q9"	"p_q13a"	"p_q13b"
##	[82]	"p_q13c"	"p_q13d"	"p_q14a"
##	[85]	"p_q14b"	"p_q14c"	"p_q14d"
##	[88]	"p_q1a_1"	"p_q1a_2"	"p_q1a_3"
##	[91]	"p_q1a_4"	"p_q1a_5"	"p_q1b_1"
##	[94]	"p_q1b_2"	"p_q1b_3"	"p_q1b_4"
##	[97]	"p_q1b_5"	"p_q1c_1"	"p_q1c_2"
##	[100]	"p_q1c_3"	"p_q1c_4"	"p_q1c_5"
##	[103]	"p_q1d_1"	"p_q1d_2"	"p_q1d_3"
##	[106]	"p_q1d_4"	"p_q1d_5"	"p_q1e_1"
##	[109]	"p_q1e_2"	"p_q1e_3"	"p_q1e_4"
##	[112]	"p_q1e_5"	"p_q1f_1"	"p_q1f_2"
##	[115]	"p_q1f_3"	"p_q1f_4"	"p_q1f_5"
##	[118]	"p_q1g_1"	"p_q1g_2"	"p_q1g_3"
##	[121]	"p_q1g_4"	"p_q1g_5"	"p_q2a_1"
##	[124]	"p_q2a_2"	"p_q2a_3"	"p_q2a_4"
##	[127]	"p_q2a_5"	"p_q2b_1"	"p_q2b_2"
##	[130]	"p_q2b_3"	"p_q2b_4"	"p_q2b_5"
##	[133]	"p_q3a_1"	"p_q3a_2"	"p_q3a_3"
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##	[142]	"p_q3b_5"	"p_q3c_1"	"p_q3c_2"
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##	[166]	"p_q4b_4"	"p_q4b_5"	"p_q4c_1"
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##	[391]	"p_N_q1b_2"	"p_N_q1b_3"	"p_N_q1b_4"
##	[394]	"p_N_q1b_5"	"p_N_q1c_1"	"p_N_q1c_2"
##	[397]	"p_N_q1c_3"	"p_N_q1c_4"	"p_N_q1c_5"
##	[400]	"p_N_q1d_1"	"p_N_q1d_2"	"p_N_q1d_3"
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##	[442]	"p_N_q3c_3"	"p_N_q3c_4"	"p_N_q3c_5"
##	[445]	"p_N_q3d_1"	"p_N_q3d_2"	"p_N_q3d_3"
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##	[460]	"p_N_q4b_1"	"p_N_q4b_2"	"p_N_q4b_3"
##	[463]	"p_N_q4b_4"	"p_N_q4b_5"	"p_N_q4c_1"
##	[466]	"p_N_q4c_2"	"p_N_q4c_3"	"p_N_q4c_4"
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##	[475]	"p_N_q5f"	"p_N_q5g"	"p_N_q5h"
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##	[481]	"p_N_q5l"	"p_N_q5m"	"p_N_q5n"
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##	[496]	"p_N_q6c_3"	"p_N_q6c_4"	"p_N_q6c_5"
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##	[568]	"p_N_q10b_1"	"p_N_q10b_2"	"p_N_q10b_3"
##	[571]	"p_N_q10b_4"	"p_N_q10b_5"	"p_N_q10c_1"
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##	[601]	"p_N_q11b_4"	"p_N_q11b_5"	"p_N_q11c_1"
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##	[607]	"p_N_q11c_5"	"p_N_q11d_1"	"p_N_q11d_2"
##	[610]	"p_N_q11d_3"	"p_N_q11d_4"	"p_N_q11d_5"
##	[613]	"p_N_q11e_1"	"p_N_q11e_2"	"p_N_q11e_3"
##	[616]	"p_N_q11e_4"	"p_N_q11e_5"	"p_N_q12a_1"
##	[619]	"p_N_q12a_2"	"p_N_q12a_3"	"p_N_q12a_4"
##	[622]	"p_N_q12b_1"	"p_N_q12b_2"	"p_N_q12b_3"
##	[625]	"p_N_q12b_4"	"p_N_q12c_1"	"p_N_q12c_2"
##	[628]	"p_N_q12c_3"	"p_N_q12c_4"	"p_N_q12d_1"
##	[631]	"p_N_q12d_2"	"p_N_q12d_3"	"p_N_q12d_4"
##	[634]	"p_N_q12e_1"	"p_N_q12e_2"	"p_N_q12e_3"
##	[637]	"p_N_q12e_4"	"p_N_q12f_1"	"p_N_q12f_2"
##	[640]	"p_N_q12f_3"	"p_N_q12f_4"	"p_N_q13a_1"
##	[643]	"p_N_q13a_2"	"p_N_q13a_3"	"p_N_q13a_4"
##	[646]	"p_N_q13a_5"	"p_N_q13b_1"	"p_N_q13b_2"
##	[649]	"p_N_q13b_3"	"p_N_q13b_4"	"p_N_q13b_5"
##	[652]	"p_N_q13c_1"	"p_N_q13c_2"	"p_N_q13c_3"
##	[655]	"p_N_q13c_4"	"p_N_q13c_5"	"p_N_q13d_1"

##	[658]	"p_N_q13d_2"	"p_N_q13d_3"	"p_N_q13d_4"
##	[661]	"p_N_q13d_5"	"p_N_q14a_1"	"p_N_q14a_2"
##	[664]	"p_N_q14a_3"	"p_N_q14a_4"	"p_N_q14a_5"
##	[667]	"p_N_q14b_1"	"p_N_q14b_2"	"p_N_q14b_3"
##	[670]	"p_N_q14b_4"	"p_N_q14b_5"	"p_N_q14c_1"
##	[673]	"p_N_q14c_2"	"p_N_q14c_3"	"p_N_q14c_4"
##	[676]	"p_N_q14c_5"	"p_N_q14d_1"	"p_N_q14d_2"
##	[679]	"p_N_q14d_3"	"p_N_q14d_4"	"p_N_q14d_5"
##	[682]	"t_q1g"	"t_q6d"	"t_q6e"
##	[685]	"t_q6f"	"t_q13a"	"t_q13b"
##	[688]	"t_q13c"	"t_q13d"	"t_q13e"
##	[691]	"t_q13f"	"t_q13g"	"t_q13h"
##	[694]	"t_q13i"	"t_q13j"	"t_q13k"
##	[697]	"t_q13l"	"t_q13m"	"t_q13n"
##	[700]	"t_q1a"	"t_q1b"	"t_q1c"
##	[703]	"t_q1f"	"t_q6h"	"t_q8c"
##	[706]	"t_q11c"	"t_q11d"	"t_q12"
##	[709]	"t_q3_1"	"t_q3_2"	"t_q4"
##	[712]	"t_q5a"	"t_q5b"	"t_q6b"
##	[715]	"t_q6c"	"t_q8a"	"t_q8b"
##	[718]	"t_q9"	"t_q10"	"t_q1d"
##	[721]	"t_q1e"	"t_q2a"	"t_q2b"
##	[724]	"t_q2c"	"t_q2d"	"t_q2e"
##	[727]	"t_q2f"	"t_q6a"	"t_q6g"
##	[730]	"t_q6i"	"t_q6j"	"t_q6k"
##	[733]	"t_q7a"	"t_q7b"	"t_q7c"
##	[736]	"t_q7d"	"t_q7e"	"t_q7f"
##	[739]	"t_q11a"	"t_q11b"	"t_q14"
##	[742]	"t_q15a"	"t_q15b"	"t_q15c"
##	[745]	"t_q15d"	"t_q16a"	"t_q16b"
##	[748]	"t_q16c"	"t_q16d"	"t_q17a"
##	[751]	"t_q17b"	"t_q17c"	"t_q17d"
##	[754]	"t_q17e"	"t_q17f"	"t_q1a_1"
##	[757]	"t_q1a_2"	"t_q1a_3"	"t_q1a_4"
##	[760]	"t_q1b_1"	"t_q1b_2"	"t_q1b_3"
##	[763]	"t_q1b_4"	"t_q1c_1"	"t_q1c_2"
##	[766]	"t_q1c_3"	"t_q1c_4"	"t_q1d_1"
##	[769]	"t_q1d_2"	"t_q1d_3"	"t_q1d_4"
##	[772]	"t_q1e_1"	"t_q1e_2"	"t_q1e_3"
##	[775]	"t_q1e_4"	"t_q1f_1"	"t_q1f_2"
##	[778]	"t_q1f_3"	"t_q1f_4"	"t_q1g_1"
##	[781]	"t_q1g_2"	"t_q1g_3"	"t_q1g_4"
##	[784]	"t_q2a_1"	"t_q2a_2"	"t_q2a_3"
##	[787]	"t_q2a_4"	"t_q2b_1"	"t_q2b_2"
##	[790]	"t_q2b_3"	"t_q2b_4"	"t_q2c_1"
##	[793]	"t_q2c_2"	"t_q2c_3"	"t_q2c_4"
##	[796]	"t_q2d_1"	"t_q2d_2"	"t_q2d_3"
##	[799]	"t_q2d_4"	"t_q2e_1"	"t_q2e_2"
##	[802]	"t_q2e_3"	"t_q2e_4"	"t_q2f_1"
##	[805]	"t_q2f_2"	"t_q2f_3"	"t_q2f_4"
##	[808]	"t_q3a_1"	"t_q3a_2"	"t_q3a_3"
##	[811]	"t_q3b_1"	"t_q3b_2"	"t_q3b_3"
##	[814]	"t_q3c_1"	"t_q3c_2"	"t_q3c_3"
##	[817]	"t_q3d_1"	"t_q3d_2"	"t_q3d_3"

##	[820]	"t_q3e_1"	"t_q3e_2"	"t_q3e_3"
##	[823]	"t_q3f_1"	"t_q3f_2"	"t_q3f_3"
##	[826]	"t_q3g_1"	"t_q3g_2"	"t_q3g_3"
##	[829]	"t_q3h_1"	"t_q3h_2"	"t_q3h_3"
##	[832]	"t_q3i_1"	"t_q3i_2"	"t_q3i_3"
##	[835]	"t_q3j_1"	"t_q3j_2"	"t_q3j_3"
##	[838]	"t_q3k_1"	"t_q3k_2"	"t_q3k_3"
##	[841]	"t_q3l_1"	"t_q3l_2"	"t_q3l_3"
##	[844]	"t_q4_1"	"t_q4_2"	"t_q4_3"
##	[847]	"t_q4_4"	"t_q5a_1"	"t_q5a_2"
##	[850]	"t_q5a_3"	"t_q5a_4"	"t_q5b_1"
##	[853]	"t_q5b_2"	"t_q5b_3"	"t_q5b_4"
##	[856]	"t_q6a_1"	"t_q6a_2"	"t_q6a_3"
##	[859]	"t_q6a_4"	"t_q6b_1"	"t_q6b_2"
##	[862]	"t_q6b_3"	"t_q6b_4"	"t_q6c_1"
##	[865]	"t_q6c_2"	"t_q6c_3"	"t_q6c_4"
##	[868]	"t_q6d_1"	"t_q6d_2"	"t_q6d_3"
##	[871]	"t_q6d_4"	"t_q6e_1"	"t_q6e_2"
##	[874]	"t_q6e_3"	"t_q6e_4"	"t_q6f_1"
##	[877]	"t_q6f_2"	"t_q6f_3"	"t_q6f_4"
##	[880]	"t_q6g_1"	"t_q6g_2"	"t_q6g_3"
##	[883]	"t_q6g_4"	"t_q6h_1"	"t_q6h_2"
##	[886]	"t_q6h_3"	"t_q6h_4"	"t_q6i_1"
##	[889]	"t_q6i_2"	"t_q6i_3"	"t_q6i_4"
##	[892]	"t_q6j_1"	"t_q6j_2"	"t_q6j_3"
##	[895]	"t_q6j_4"	"t_q6k_1"	"t_q6k_2"
##	[898]	"t_q6k_3"	"t_q6k_4"	"t_q7a_1"
##	[901]	"t_q7a_2"	"t_q7a_3"	"t_q7a_4"
##	[904]	"t_q7a_5"	"t_q7b_1"	"t_q7b_2"
##	[907]	"t_q7b_3"	"t_q7b_4"	"t_q7b_5"
##	[910]	"t_q7c_1"	"t_q7c_2"	"t_q7c_3"
##	[913]	"t_q7c_4"	"t_q7c_5"	"t_q7d_1"
##	[916]	"t_q7d_2"	"t_q7d_3"	"t_q7d_4"
##	[919]	"t_q7d_5"	"t_q7e_1"	"t_q7e_2"
##	[922]	"t_q7e_3"	"t_q7e_4"	"t_q7e_5"
##	[925]	"t_q7f_1"	"t_q7f_2"	"t_q7f_3"
##	[928]	"t_q7f_4"	"t_q7f_5"	"t_q8a_1"
##	[931]	"t_q8a_2"	"t_q8a_3"	"t_q8a_4"
##	[934]	"t_q8b_1"	"t_q8b_2"	"t_q8b_3"
##	[937]	"t_q8b_4"	"t_q8c_1"	"t_q8c_2"
##	[940]	"t_q8c_3"	"t_q8c_4"	"t_q9_1"
##	[943]	"t_q9_2"	"t_q9_3"	"t_q9_4"
##	[946]	"t_q9_5"	"t_q10_1"	"t_q10_2"
##	[949]	"t_q10_3"	"t_q10_4"	"t_q10_5"
##	[952]	"t_q11a_1"	"t_q11a_2"	"t_q11a_3"
##	[955]	"t_q11a_4"	"t_q11a_5"	"t_q11b_1"
##	[958]	"t_q11b_2"	"t_q11b_3"	"t_q11b_4"
##	[961]	"t_q11b_5"	"t_q11c_1"	"t_q11c_2"
##	[964]	"t_q11c_3"	"t_q11c_4"	"t_q11c_5"
##	[967]	"t_q11d_1"	"t_q11d_2"	"t_q11d_3"
##	[970]	"t_q11d_4"	"t_q11d_5"	"t_q12_1"
##	[973]	"t_q12_2"	"t_q12_3"	"t_q12_4"
##	[976]	"t_q12_5"	"t_q12_6"	"t_q13a_1"
##	[979]	"t_q13a_2"	"t_q13a_3"	"t_q13a_4"

## [982]	"t_q13b_1"	"t_q13b_2"	"t_q13b_3"
## [985]	"t_q13b_4"	"t_q13c_1"	"t_q13c_2"
## [988]	"t_q13c_3"	"t_q13c_4"	"t_q13d_1"
## [991]	"t_q13d_2"	"t_q13d_3"	"t_q13d_4"
## [994]	"t_q13e_1"	"t_q13e_2"	"t_q13e_3"
## [997]	"t_q13e_4"	"t_q13f_1"	"t_q13f_2"
## [1000]	"t_q13f_3"	"t_q13f_4"	"t_q13g_1"
## [1003]	"t_q13g_2"	"t_q13g_3"	"t_q13g_4"
## [1006]	"t_q13h_1"	"t_q13h_2"	"t_q13h_3"
## [1009]	"t_q13h_4"	"t_q13i_1"	"t_q13i_2"
## [1012]	"t_q13i_3"	"t_q13i_4"	"t_q13j_1"
## [1015]	"t_q13j_2"	"t_q13j_3"	"t_q13j_4"
## [1018]	"t_q13k_1"	"t_q13k_2"	"t_q13k_3"
## [1021]	"t_q13k_4"	"t_q13l_1"	"t_q13l_2"
## [1024]	"t_q13l_3"	"t_q13l_4"	"t_q13m_1"
## [1027]	"t_q13m_2"	"t_q13m_3"	"t_q13m_4"
## [1030]	"t_q13n_1"	"t_q13n_2"	"t_q13n_3"
## [1033]	"t_q13n_4"	"t_q14_1"	"t_q14_2"
## [1036]	"t_q14_3"	"t_q14_4"	"t_q14_5"
## [1039]	"t_q14_6"	"t_q15a_1"	"t_q15a_2"
## [1042]	"t_q15a_3"	"t_q15a_4"	"t_q15a_5"
## [1045]	"t_q15b_1"	"t_q15b_2"	"t_q15b_3"
## [1048]	"t_q15b_4"	"t_q15b_5"	"t_q15c_1"
## [1051]	"t_q15c_2"	"t_q15c_3"	"t_q15c_4"
## [1054]	"t_q15c_5"	"t_q15d_1"	"t_q15d_2"
## [1057]	"t_q15d_3"	"t_q15d_4"	"t_q15d_5"
## [1060]	"t_q16a_1"	"t_q16a_2"	"t_q16a_3"
## [1063]	"t_q16a_4"	"t_q16a_5"	"t_q16b_1"
## [1066]	"t_q16b_2"	"t_q16b_3"	"t_q16b_4"
## [1069]	"t_q16b_5"	"t_q16c_1"	"t_q16c_2"
## [1072]	"t_q16c_3"	"t_q16c_4"	"t_q16c_5"
## [1075]	"t_q16d_1"	"t_q16d_2"	"t_q16d_3"
## [1078]	"t_q16d_4"	"t_q16d_5"	"t_q17a_1"
## [1081]	"t_q17a_2"	"t_q17a_3"	"t_q17a_4"
## [1084]	"t_q17b_1"	"t_q17b_2"	"t_q17b_3"
## [1087]	"t_q17b_4"	"t_q17c_1"	"t_q17c_2"
## [1090]	"t_q17c_3"	"t_q17c_4"	"t_q17d_1"
## [1093]	"t_q17d_2"	"t_q17d_3"	"t_q17d_4"
## [1096]	"t_q17e_1"	"t_q17e_2"	"t_q17e_3"
## [1099]	"t_q17e_4"	"t_q17f_1"	"t_q17f_2"
## [1102]	"t_q17f_3"	"t_q17f_4"	"t_N_q1a_1"
## [1105]	"t_N_q1a_2"	"t_N_q1a_3"	"t_N_q1a_4"
## [1108]	"t_N_q1b_1"	"t_N_q1b_2"	"t_N_q1b_3"
## [1111]	"t_N_q1b_4"	"t_N_q1c_1"	"t_N_q1c_2"
## [1114]	"t_N_q1c_3"	"t_N_q1c_4"	"t_N_q1d_1"
## [1117]	"t_N_q1d_2"	"t_N_q1d_3"	"t_N_q1d_4"
## [1120]	"t_N_q1e_1"	"t_N_q1e_2"	"t_N_q1e_3"
## [1123]	"t_N_q1e_4"	"t_N_q1f_1"	"t_N_q1f_2"
## [1126]	"t_N_q1f_3"	"t_N_q1f_4"	"t_N_q1g_1"
## [1129]	"t_N_q1g_2"	"t_N_q1g_3"	"t_N_q1g_4"
## [1132]	"t_N_q2a_1"	"t_N_q2a_2"	"t_N_q2a_3"
## [1135]	"t_N_q2a_4"	"t_N_q2b_1"	"t_N_q2b_2"
## [1138]	"t_N_q2b_3"	"t_N_q2b_4"	"t_N_q2c_1"
## [1141]	"t_N_q2c_2"	"t_N_q2c_3"	"t_N_q2c_4"

## [1144]	"t_N_q2d_1"	"t_N_q2d_2"	"t_N_q2d_3"
## [1147]	"t_N_q2d_4"	"t_N_q2e_1"	"t_N_q2e_2"
## [1150]	"t_N_q2e_3"	"t_N_q2e_4"	"t_N_q2f_1"
## [1153]	"t_N_q2f_2"	"t_N_q2f_3"	"t_N_q2f_4"
## [1156]	"t_N_q3a_1"	"t_N_q3a_2"	"t_N_q3a_3"
## [1159]	"t_N_q3b_1"	"t_N_q3b_2"	"t_N_q3b_3"
## [1162]	"t_N_q3c_1"	"t_N_q3c_2"	"t_N_q3c_3"
## [1165]	"t_N_q3d_1"	"t_N_q3d_2"	"t_N_q3d_3"
## [1168]	"t_N_q3e_1"	"t_N_q3e_2"	"t_N_q3e_3"
## [1171]	"t_N_q3f_1"	"t_N_q3f_2"	"t_N_q3f_3"
## [1174]	"t_N_q3g_1"	"t_N_q3g_2"	"t_N_q3g_3"
## [1177]	"t_N_q3h_1"	"t_N_q3h_2"	"t_N_q3h_3"
## [1180]	"t_N_q3i_1"	"t_N_q3i_2"	"t_N_q3i_3"
## [1183]	"t_N_q3j_1"	"t_N_q3j_2"	"t_N_q3j_3"
## [1186]	"t_N_q3k_1"	"t_N_q3k_2"	"t_N_q3k_3"
## [1189]	"t_N_q3l_1"	"t_N_q3l_2"	"t_N_q3l_3"
## [1192]	"t_N_q4_1"	"t_N_q4_2"	"t_N_q4_3"
## [1195]	"t_N_q4_4"	"t_N_q5a_1"	"t_N_q5a_2"
## [1198]	"t_N_q5a_3"	"t_N_q5a_4"	"t_N_q5b_1"
## [1201]	"t_N_q5b_2"	"t_N_q5b_3"	"t_N_q5b_4"
## [1204]	"t_N_q6a_1"	"t_N_q6a_2"	"t_N_q6a_3"
## [1207]	"t_N_q6a_4"	"t_N_q6b_1"	"t_N_q6b_2"
## [1210]	"t_N_q6b_3"	"t_N_q6b_4"	"t_N_q6c_1"
## [1213]	"t_N_q6c_2"	"t_N_q6c_3"	"t_N_q6c_4"
## [1216]	"t_N_q6d_1"	"t_N_q6d_2"	"t_N_q6d_3"
## [1219]	"t_N_q6d_4"	"t_N_q6e_1"	"t_N_q6e_2"
## [1222]	"t_N_q6e_3"	"t_N_q6e_4"	"t_N_q6f_1"
## [1225]	"t_N_q6f_2"	"t_N_q6f_3"	"t_N_q6f_4"
## [1228]	"t_N_q6g_1"	"t_N_q6g_2"	"t_N_q6g_3"
## [1231]	"t_N_q6g_4"	"t_N_q6h_1"	"t_N_q6h_2"
## [1234]	"t_N_q6h_3"	"t_N_q6h_4"	"t_N_q6i_1"
## [1237]	"t_N_q6i_2"	"t_N_q6i_3"	"t_N_q6i_4"
## [1240]	"t_N_q6j_1"	"t_N_q6j_2"	"t_N_q6j_3"
## [1243]	"t_N_q6j_4"	"t_N_q6k_1"	"t_N_q6k_2"
## [1246]	"t_N_q6k_3"	"t_N_q6k_4"	"t_N_q7a_1"
## [1249]	"t_N_q7a_2"	"t_N_q7a_3"	"t_N_q7a_4"
## [1252]	"t_N_q7a_5"	"t_N_q7b_1"	"t_N_q7b_2"
## [1255]	"t_N_q7b_3"	"t_N_q7b_4"	"t_N_q7b_5"
## [1258]	"t_N_q7c_1"	"t_N_q7c_2"	"t_N_q7c_3"
## [1261]	"t_N_q7c_4"	"t_N_q7c_5"	"t_N_q7d_1"
## [1264]	"t_N_q7d_2"	"t_N_q7d_3"	"t_N_q7d_4"
## [1267]	"t_N_q7d_5"	"t_N_q7e_1"	"t_N_q7e_2"
## [1270]	"t_N_q7e_3"	"t_N_q7e_4"	"t_N_q7e_5"
## [1273]	"t_N_q7f_1"	"t_N_q7f_2"	"t_N_q7f_3"
## [1276]	"t_N_q7f_4"	"t_N_q7f_5"	"t_N_q8a_1"
## [1279]	"t_N_q8a_2"	"t_N_q8a_3"	"t_N_q8a_4"
## [1282]	"t_N_q8b_1"	"t_N_q8b_2"	"t_N_q8b_3"
## [1285]	"t_N_q8b_4"	"t_N_q8c_1"	"t_N_q8c_2"
## [1288]	"t_N_q8c_3"	"t_N_q8c_4"	"t_N_q9_1"
## [1291]	"t_N_q9_2"	"t_N_q9_3"	"t_N_q9_4"
## [1294]	"t_N_q9_5"	"t_N_q10_1"	"t_N_q10_2"
## [1297]	"t_N_q10_3"	"t_N_q10_4"	"t_N_q10_5"
## [1300]	"t_N_q11a_1"	"t_N_q11a_2"	"t_N_q11a_3"
## [1303]	"t_N_q11a_4"	"t_N_q11a_5"	"t_N_q11b_1"

## [1306]	"t_N_q11b_2"	"t_N_q11b_3"	"t_N_q11b_4"
## [1309]	"t_N_q11b_5"	"t_N_q11c_1"	"t_N_q11c_2"
## [1312]	"t_N_q11c_3"	"t_N_q11c_4"	"t_N_q11c_5"
## [1315]	"t_N_q11d_1"	"t_N_q11d_2"	"t_N_q11d_3"
## [1318]	"t_N_q11d_4"	"t_N_q11d_5"	"t_N_q12_1"
## [1321]	"t_N_q12_2"	"t_N_q12_3"	"t_N_q12_4"
## [1324]	"t_N_q12_5"	"t_N_q12_6"	"t_N_q13a_1"
## [1327]	"t_N_q13a_2"	"t_N_q13a_3"	"t_N_q13a_4"
## [1330]	"t_N_q13b_1"	"t_N_q13b_2"	"t_N_q13b_3"
## [1333]	"t_N_q13b_4"	"t_N_q13c_1"	"t_N_q13c_2"
## [1336]	"t_N_q13c_3"	"t_N_q13c_4"	"t_N_q13d_1"
## [1339]	"t_N_q13d_2"	"t_N_q13d_3"	"t_N_q13d_4"
## [1342]	"t_N_q13e_1"	"t_N_q13e_2"	"t_N_q13e_3"
## [1345]	"t_N_q13e_4"	"t_N_q13f_1"	"t_N_q13f_2"
## [1348]	"t_N_q13f_3"	"t_N_q13f_4"	"t_N_q13g_1"
## [1351]	"t_N_q13g_2"	"t_N_q13g_3"	"t_N_q13g_4"
## [1354]	"t_N_q13h_1"	"t_N_q13h_2"	"t_N_q13h_3"
## [1357]	"t_N_q13h_4"	"t_N_q13i_1"	"t_N_q13i_2"
## [1360]	"t_N_q13i_3"	"t_N_q13i_4"	"t_N_q13j_1"
## [1363]	"t_N_q13j_2"	"t_N_q13j_3"	"t_N_q13j_4"
## [1366]	"t_N_q13k_1"	"t_N_q13k_2"	"t_N_q13k_3"
## [1369]	"t_N_q13k_4"	"t_N_q13l_1"	"t_N_q13l_2"
## [1372]	"t_N_q13l_3"	"t_N_q13l_4"	"t_N_q13m_1"
## [1375]	"t_N_q13m_2"	"t_N_q13m_3"	"t_N_q13m_4"
## [1378]	"t_N_q13n_1"	"t_N_q13n_2"	"t_N_q13n_3"
## [1381]	"t_N_q13n_4"	"t_N_q14_1"	"t_N_q14_2"
## [1384]	"t_N_q14_3"	"t_N_q14_4"	"t_N_q14_5"
## [1387]	"t_N_q14_6"	"t_N_q15a_1"	"t_N_q15a_2"
## [1390]	"t_N_q15a_3"	"t_N_q15a_4"	"t_N_q15a_5"
## [1393]	"t_N_q15b_1"	"t_N_q15b_2"	"t_N_q15b_3"
## [1396]	"t_N_q15b_4"	"t_N_q15b_5"	"t_N_q15c_1"
## [1399]	"t_N_q15c_2"	"t_N_q15c_3"	"t_N_q15c_4"
## [1402]	"t_N_q15c_5"	"t_N_q15d_1"	"t_N_q15d_2"
## [1405]	"t_N_q15d_3"	"t_N_q15d_4"	"t_N_q15d_5"
## [1408]	"t_N_q16a_1"	"t_N_q16a_2"	"t_N_q16a_3"
## [1411]	"t_N_q16a_4"	"t_N_q16a_5"	"t_N_q16b_1"
## [1414]	"t_N_q16b_2"	"t_N_q16b_3"	"t_N_q16b_4"
## [1417]	"t_N_q16b_5"	"t_N_q16c_1"	"t_N_q16c_2"
## [1420]	"t_N_q16c_3"	"t_N_q16c_4"	"t_N_q16c_5"
## [1423]	"t_N_q16d_1"	"t_N_q16d_2"	"t_N_q16d_3"
## [1426]	"t_N_q16d_4"	"t_N_q16d_5"	"t_N_q17a_1"
## [1429]	"t_N_q17a_2"	"t_N_q17a_3"	"t_N_q17a_4"
## [1432]	"t_N_q17b_1"	"t_N_q17b_2"	"t_N_q17b_3"
## [1435]	"t_N_q17b_4"	"t_N_q17c_1"	"t_N_q17c_2"
## [1438]	"t_N_q17c_3"	"t_N_q17c_4"	"t_N_q17d_1"
## [1441]	"t_N_q17d_2"	"t_N_q17d_3"	"t_N_q17d_4"
## [1444]	"t_N_q17e_1"	"t_N_q17e_2"	"t_N_q17e_3"
## [1447]	"t_N_q17e_4"	"t_N_q17f_1"	"t_N_q17f_2"
## [1450]	"t_N_q17f_3"	"t_N_q17f_4"	"s_q5a"
## [1453]	"s_q5b"	"s_q5c"	"s_q11a"
## [1456]	"s_q11b"	"s_q11c"	"s_q12a"
## [1459]	"s_q12b"	"s_q12c"	"s_q12d"
## [1462]	"s_q12e"	"s_q12f"	"s_q12g"
## [1465]	"s_q13a"	"s_q13b"	"s_q13c"

## [1468]	"s_q13d"	"s_q13e"	"s_q13f"
## [1471]	"s_q13g"	"s_q1b"	"s_q3a"
## [1474]	"s_q3b"	"s_q7b"	"s_q7c"
## [1477]	"s_q7d"	"s_q1a"	"s_q1c"
## [1480]	"s_q4a"	"s_q4b"	"s_q5d"
## [1483]	"s_q5e"	"s_q5g"	"s_q8"
## [1486]	"s_q9"	"s_q10"	"s_q2a"
## [1489]	"s_q2b"	"s_q2c"	"s_q2d"
## [1492]	"s_q2e"	"s_q2f"	"s_q2g"
## [1495]	"s_q5f"	"s_q6a"	"s_q6b"
## [1498]	"s_q7a"	"s_q14"	"s_q1a_1"
## [1501]	"s_q1a_2"	"s_q1a_3"	"s_q1a_4"
## [1504]	"s_q1b_1"	"s_q1b_2"	"s_q1b_3"
## [1507]	"s_q1b_4"	"s_q1c_1"	"s_q1c_2"
## [1510]	"s_q1c_3"	"s_q1c_4"	"s_q2a_1"
## [1513]	"s_q2a_2"	"s_q2a_3"	"s_q2a_4"
## [1516]	"s_q2b_1"	"s_q2b_2"	"s_q2b_3"
## [1519]	"s_q2b_4"	"s_q2c_1"	"s_q2c_2"
## [1522]	"s_q2c_3"	"s_q2c_4"	"s_q2d_1"
## [1525]	"s_q2d_2"	"s_q2d_3"	"s_q2d_4"
## [1528]	"s_q2e_1"	"s_q2e_2"	"s_q2e_3"
## [1531]	"s_q2e_4"	"s_q2f_1"	"s_q2f_2"
## [1534]	"s_q2f_3"	"s_q2f_4"	"s_q2g_1"
## [1537]	"s_q2g_2"	"s_q2g_3"	"s_q2g_4"
## [1540]	"s_q3a_1"	"s_q3a_2"	"s_q3a_3"
## [1543]	"s_q3a_4"	"s_q3b_1"	"s_q3b_2"
## [1546]	"s_q3b_3"	"s_q3b_4"	"s_q4a_1"
## [1549]	"s_q4a_2"	"s_q4a_3"	"s_q4a_4"
## [1552]	"s_q4b_1"	"s_q4b_2"	"s_q4b_3"
## [1555]	"s_q4b_4"	"s_q5a_1"	"s_q5a_2"
## [1558]	"s_q5a_3"	"s_q5a_4"	"s_q5a_5"
## [1561]	"s_q5b_1"	"s_q5b_2"	"s_q5b_3"
## [1564]	"s_q5b_4"	"s_q5b_5"	"s_q5c_1"
## [1567]	"s_q5c_2"	"s_q5c_3"	"s_q5c_4"
## [1570]	"s_q5c_5"	"s_q5d_1"	"s_q5d_2"
## [1573]	"s_q5d_3"	"s_q5d_4"	"s_q5d_5"
## [1576]	"s_q5e_1"	"s_q5e_2"	"s_q5e_3"
## [1579]	"s_q5e_4"	"s_q5e_5"	"s_q5f_1"
## [1582]	"s_q5f_2"	"s_q5f_3"	"s_q5f_4"
## [1585]	"s_q5f_5"	"s_q5g_1"	"s_q5g_2"
## [1588]	"s_q5g_3"	"s_q5g_4"	"s_q5g_5"
## [1591]	"s_q6a_1"	"s_q6a_2"	"s_q6a_3"
## [1594]	"s_q6a_4"	"s_q6b_1"	"s_q6b_2"
## [1597]	"s_q6b_3"	"s_q6b_4"	"s_q7a_1"
## [1600]	"s_q7a_2"	"s_q7a_3"	"s_q7a_4"
## [1603]	"s_q7a_5"	"s_q7b_1"	"s_q7b_2"
## [1606]	"s_q7b_3"	"s_q7b_4"	"s_q7b_5"
## [1609]	"s_q7c_1"	"s_q7c_2"	"s_q7c_3"
## [1612]	"s_q7c_4"	"s_q7c_5"	"s_q7d_1"
## [1615]	"s_q7d_2"	"s_q7d_3"	"s_q7d_4"
## [1618]	"s_q7d_5"	"s_q8a_1"	"s_q8a_2"
## [1621]	"s_q8a_3"	"s_q8b_1"	"s_q8b_2"
## [1624]	"s_q8b_3"	"s_q8c_1"	"s_q8c_2"
## [1627]	"s_q8c_3"	"s_q8d_1"	"s_q8d_2"

```
## [1630] "s_q8d_3"          "s_q8e_1"          "s_q8e_2"
## [1633] "s_q8e_3"          "s_q8f_1"          "s_q8f_2"
## [1636] "s_q8f_3"          "s_q8g_1"          "s_q8g_2"
## [1639] "s_q8g_3"          "s_q8h_1"          "s_q8h_2"
## [1642] "s_q8h_3"          "s_q8i_1"          "s_q8i_2"
## [1645] "s_q8i_3"          "s_q8j_1"          "s_q8j_2"
## [1648] "s_q8j_3"          "s_q8k_1"          "s_q8k_2"
## [1651] "s_q8k_3"          "s_q8l_1"          "s_q8l_2"
## [1654] "s_q8l_3"          "s_q9a_1"          "s_q9a_2"
## [1657] "s_q9a_3"          "s_q9b_1"          "s_q9b_2"
## [1660] "s_q9b_3"          "s_q9c_1"          "s_q9c_2"
## [1663] "s_q9c_3"          "s_q9d_1"          "s_q9d_2"
## [1666] "s_q9d_3"          "s_q9e_1"          "s_q9e_2"
## [1669] "s_q9e_3"          "s_q9f_1"          "s_q9f_2"
## [1672] "s_q9f_3"          "s_q9g_1"          "s_q9g_2"
## [1675] "s_q9g_3"          "s_q9h_1"          "s_q9h_2"
## [1678] "s_q9h_3"          "s_q9i_1"          "s_q9i_2"
## [1681] "s_q9i_3"          "s_q9j_1"          "s_q9j_2"
## [1684] "s_q9j_3"          "s_q9k_1"          "s_q9k_2"
## [1687] "s_q9k_3"          "s_q9l_1"          "s_q9l_2"
## [1690] "s_q9l_3"          "s_q10_1"          "s_q10_2"
## [1693] "s_q10_3"          "s_q10_4"          "s_q11a_1"
## [1696] "s_q11a_2"          "s_q11a_3"          "s_q11a_4"
## [1699] "s_q11b_1"          "s_q11b_2"          "s_q11b_3"
## [1702] "s_q11b_4"          "s_q11c_1"          "s_q11c_2"
## [1705] "s_q11c_3"          "s_q11c_4"          "s_q12a_1"
## [1708] "s_q12a_2"          "s_q12a_3"          "s_q12a_4"
## [1711] "s_q12b_1"          "s_q12b_2"          "s_q12b_3"
## [1714] "s_q12b_4"          "s_q12c_1"          "s_q12c_2"
## [1717] "s_q12c_3"          "s_q12c_4"          "s_q12d_1"
## [1720] "s_q12d_2"          "s_q12d_3"          "s_q12d_4"
## [1723] "s_q12e_1"          "s_q12e_2"          "s_q12e_3"
## [1726] "s_q12e_4"          "s_q12f_1"          "s_q12f_2"
## [1729] "s_q12f_3"          "s_q12f_4"          "s_q12g_1"
## [1732] "s_q12g_2"          "s_q12g_3"          "s_q12g_4"
## [1735] "s_q13a_1"          "s_q13a_2"          "s_q13a_3"
## [1738] "s_q13a_4"          "s_q13b_1"          "s_q13b_2"
## [1741] "s_q13b_3"          "s_q13b_4"          "s_q13c_1"
## [1744] "s_q13c_2"          "s_q13c_3"          "s_q13c_4"
## [1747] "s_q13d_1"          "s_q13d_2"          "s_q13d_3"
## [1750] "s_q13d_4"          "s_q13e_1"          "s_q13e_2"
## [1753] "s_q13e_3"          "s_q13e_4"          "s_q13f_1"
## [1756] "s_q13f_2"          "s_q13f_3"          "s_q13f_4"
## [1759] "s_q13g_1"          "s_q13g_2"          "s_q13g_3"
## [1762] "s_q13g_4"          "s_q14_1"          "s_q14_2"
## [1765] "s_q14_3"          "s_q14_4"          "s_q14_5"
## [1768] "s_q14_6"          "s_q14_7"          "s_q14_8"
## [1771] "s_q14_9"          "s_q14_10"         "s_q14_11"
```

```
df_gened <- read_tsv("masterfile11_gened_final.txt")
```

```
## Rows: 1646 Columns: 1942
```

```
## -- Column specification -----
```

```
## Delimiter: "\t"
```

```
## chr      (5): dbn, bn, schoolname, studentssurveyed, schooltype
```

```
## dbl (1904): d75, highschool, rr_s, rr_t, rr_p, N_s, N_t, N_p, nr_s, nr_t, nr...
## lgl (33): p_q1, p_q3d, p_q9, p_q10, p_q12aa, p_q12ab, p_q12ac, p_q12ad, p...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
names(df_gened)
```

```
##      [1] "dbn"          "bn"           "schoolname"
##      [4] "d75"          "studentssurveyed" "highschool"
##      [7] "schooltype"   "rr_s"         "rr_t"
##     [10] "rr_p"         "N_s"          "N_t"
##     [13] "N_p"         "nr_s"         "nr_t"
##     [16] "nr_p"         "saf_p_11"     "com_p_11"
##     [19] "eng_p_11"     "aca_p_11"     "saf_t_11"
##     [22] "com_t_11"     "eng_t_11"     "aca_t_11"
##     [25] "saf_s_11"     "com_s_11"     "eng_s_11"
##     [28] "aca_s_11"     "saf_tot_11"   "com_tot_11"
##     [31] "eng_tot_11"   "aca_tot_11"   "p_q2h"
##     [34] "p_q7a"        "p_q7b"        "p_q7c"
##     [37] "p_q7d"        "p_q8a"        "p_q8b"
##     [40] "p_q8c"        "p_q8d"        "p_q8e"
##     [43] "p_q8f"        "p_q2b"        "p_q2d"
##     [46] "p_q2e"        "p_q2f"        "p_q2g"
##     [49] "p_q3a"        "p_q3b"        "p_q4b"
##     [52] "p_q4c"        "p_q11c"       "p_q2a"
##     [55] "p_q2c"        "p_q3c"        "p_q6a"
##     [58] "p_q6b"        "p_q11d"       "p_q11e"
##     [61] "p_q5"         "p_q4a"        "p_q4d"
##     [64] "p_q4e"        "p_q11a"       "p_q11b"
##     [67] "p_q11f"       "p_q1"         "p_q3d"
##     [70] "p_q9"         "p_q10"        "p_q12aa"
##     [73] "p_q12ab"      "p_q12ac"      "p_q12ad"
##     [76] "p_q12ba"      "p_q12bb"      "p_q12bc"
##     [79] "p_q12bd"      "p_q1_1"       "p_q1_2"
##     [82] "p_q1_3"       "p_q1_4"       "p_q1_5"
##     [85] "p_q1_6"       "p_q1_7"       "p_q1_8"
##     [88] "p_q1_9"       "p_q1_10"      "p_q1_11"
##     [91] "p_q1_12"      "p_q1_13"      "p_q1_14"
##     [94] "p_q2a_1"      "p_q2a_2"      "p_q2a_3"
##     [97] "p_q2a_4"      "p_q2a_5"      "p_q2b_1"
##    [100] "p_q2b_2"      "p_q2b_3"      "p_q2b_4"
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##    [106] "p_q2c_3"      "p_q2c_4"      "p_q2c_5"
##    [109] "p_q2d_1"      "p_q2d_2"      "p_q2d_3"
##    [112] "p_q2d_4"      "p_q2d_5"      "p_q2e_1"
##    [115] "p_q2e_2"      "p_q2e_3"      "p_q2e_4"
##    [118] "p_q2e_5"      "p_q2f_1"      "p_q2f_2"
##    [121] "p_q2f_3"      "p_q2f_4"      "p_q2f_5"
##    [124] "p_q2g_1"      "p_q2g_2"      "p_q2g_3"
##    [127] "p_q2g_4"      "p_q2g_5"      "p_q2h_1"
##    [130] "p_q2h_2"      "p_q2h_3"      "p_q2h_4"
##    [133] "p_q2h_5"      "p_q3a_1"      "p_q3a_2"
##    [136] "p_q3a_3"      "p_q3a_4"      "p_q3a_5"
##    [139] "p_q3b_1"      "p_q3b_2"      "p_q3b_3"
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##	[154]	"p_q4a_1"	"p_q4a_2"	"p_q4a_3"
##	[157]	"p_q4a_4"	"p_q4a_5"	"p_q4b_1"
##	[160]	"p_q4b_2"	"p_q4b_3"	"p_q4b_4"
##	[163]	"p_q4b_5"	"p_q4c_1"	"p_q4c_2"
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##	[169]	"p_q4d_1"	"p_q4d_2"	"p_q4d_3"
##	[172]	"p_q4d_4"	"p_q4d_5"	"p_q4e_1"
##	[175]	"p_q4e_2"	"p_q4e_3"	"p_q4e_4"
##	[178]	"p_q4e_5"	"p_q5a"	"p_q5b"
##	[181]	"p_q5c"	"p_q5d"	"p_q5e"
##	[184]	"p_q5f"	"p_q5g"	"p_q5h"
##	[187]	"p_q5i"	"p_q5j"	"p_q6a_1"
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##	[235]	"p_q8d_2"	"p_q8d_3"	"p_q8d_4"
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##	[244]	"p_q8f_1"	"p_q8f_2"	"p_q8f_3"
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##	[268]	"p_q10j"	"p_q10k"	"p_q10l"
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##	[298]	"p_q12aa_4"	"p_q12aa_5"	"p_q12ab_1"
##	[301]	"p_q12ab_2"	"p_q12ab_3"	"p_q12ab_4"

##	[304]	"p_q12ab_5"	"p_q12ac_1"	"p_q12ac_2"
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##	[310]	"p_q12ad_1"	"p_q12ad_2"	"p_q12ad_3"
##	[313]	"p_q12ad_4"	"p_q12ad_5"	"p_q12ba_1"
##	[316]	"p_q12ba_2"	"p_q12ba_3"	"p_q12ba_4"
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##	[325]	"p_q12bc_1"	"p_q12bc_2"	"p_q12bc_3"
##	[328]	"p_q12bc_4"	"p_q12bc_5"	"p_q12bd_1"
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##	[343]	"p_N_q1_9"	"p_N_q1_10"	"p_N_q1_11"
##	[346]	"p_N_q1_12"	"p_N_q1_13"	"p_N_q1_14"
##	[349]	"p_N_q2a_1"	"p_N_q2a_2"	"p_N_q2a_3"
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##	[361]	"p_N_q2c_3"	"p_N_q2c_4"	"p_N_q2c_5"
##	[364]	"p_N_q2d_1"	"p_N_q2d_2"	"p_N_q2d_3"
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##	[382]	"p_N_q2g_4"	"p_N_q2g_5"	"p_N_q2h_1"
##	[385]	"p_N_q2h_2"	"p_N_q2h_3"	"p_N_q2h_4"
##	[388]	"p_N_q2h_5"	"p_N_q3a_1"	"p_N_q3a_2"
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##	[439]	"p_N_q5f"	"p_N_q5g"	"p_N_q5h"
##	[442]	"p_N_q5i"	"p_N_q5j"	"p_N_q6a_1"
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##	[457]	"p_N_q7a_4"	"p_N_q7a_5"	"p_N_q7b_1"
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##	[481]	"p_N_q8b_3"	"p_N_q8b_4"	"p_N_q8b_5"
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##	[517]	"p_N_q10d"	"p_N_q10e"	"p_N_q10f"
##	[520]	"p_N_q10g"	"p_N_q10h"	"p_N_q10i"
##	[523]	"p_N_q10j"	"p_N_q10k"	"p_N_q10l"
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##	[595]	"t_q11c"	"t_q11d"	"t_q11e"
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##	[601]	"t_q11j"	"t_q11k"	"t_q11l"
##	[604]	"t_q11m"	"t_q11n"	"t_q11o"
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##	[610]	"t_q1f"	"t_q1g"	"t_q6h"
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##	[619]	"t_q5b"	"t_q5c"	"t_q6b"
##	[622]	"t_q6c"	"t_q7a"	"t_q7d"
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##	[733]	"t_q3f_1"	"t_q3f_2"	"t_q3f_3"
##	[736]	"t_q3g_1"	"t_q3g_2"	"t_q3g_3"
##	[739]	"t_q3h_1"	"t_q3h_2"	"t_q3h_3"
##	[742]	"t_q3i_1"	"t_q3i_2"	"t_q3i_3"
##	[745]	"t_q3j_1"	"t_q3j_2"	"t_q3j_3"
##	[748]	"t_q4_1"	"t_q4_2"	"t_q4_3"
##	[751]	"t_q4_4"	"t_q5a_1"	"t_q5a_2"
##	[754]	"t_q5a_3"	"t_q5a_4"	"t_q5a_5"
##	[757]	"t_q5b_1"	"t_q5b_2"	"t_q5b_3"
##	[760]	"t_q5b_4"	"t_q5b_5"	"t_q5c_1"
##	[763]	"t_q5c_2"	"t_q5c_3"	"t_q5c_4"
##	[766]	"t_q5c_5"	"t_q6a_1"	"t_q6a_2"
##	[769]	"t_q6a_3"	"t_q6a_4"	"t_q6b_1"
##	[772]	"t_q6b_2"	"t_q6b_3"	"t_q6b_4"
##	[775]	"t_q6c_1"	"t_q6c_2"	"t_q6c_3"
##	[778]	"t_q6c_4"	"t_q6d_1"	"t_q6d_2"
##	[781]	"t_q6d_3"	"t_q6d_4"	"t_q6e_1"
##	[784]	"t_q6e_2"	"t_q6e_3"	"t_q6e_4"
##	[787]	"t_q6f_1"	"t_q6f_2"	"t_q6f_3"

##	[790]	"t_q6f_4"	"t_q6g_1"	"t_q6g_2"
##	[793]	"t_q6g_3"	"t_q6g_4"	"t_q6h_1"
##	[796]	"t_q6h_2"	"t_q6h_3"	"t_q6h_4"
##	[799]	"t_q6i_1"	"t_q6i_2"	"t_q6i_3"
##	[802]	"t_q6i_4"	"t_q6j_1"	"t_q6j_2"
##	[805]	"t_q6j_3"	"t_q6j_4"	"t_q6k_1"
##	[808]	"t_q6k_2"	"t_q6k_3"	"t_q6k_4"
##	[811]	"t_q6l_1"	"t_q6l_2"	"t_q6l_3"
##	[814]	"t_q6l_4"	"t_q6m_1"	"t_q6m_2"
##	[817]	"t_q6m_3"	"t_q6m_4"	"t_q7a_1"
##	[820]	"t_q7a_2"	"t_q7a_3"	"t_q7a_4"
##	[823]	"t_q7a_5"	"t_q7b_1"	"t_q7b_2"
##	[826]	"t_q7b_3"	"t_q7b_4"	"t_q7b_5"
##	[829]	"t_q7c_1"	"t_q7c_2"	"t_q7c_3"
##	[832]	"t_q7c_4"	"t_q7c_5"	"t_q7d_1"
##	[835]	"t_q7d_2"	"t_q7d_3"	"t_q7d_4"
##	[838]	"t_q7d_5"	"t_q7e_1"	"t_q7e_2"
##	[841]	"t_q7e_3"	"t_q7e_4"	"t_q7e_5"
##	[844]	"t_q8a_1"	"t_q8a_2"	"t_q8a_3"
##	[847]	"t_q8a_4"	"t_q8b_1"	"t_q8b_2"
##	[850]	"t_q8b_3"	"t_q8b_4"	"t_q8c_1"
##	[853]	"t_q8c_2"	"t_q8c_3"	"t_q8c_4"
##	[856]	"t_q9_1"	"t_q9_2"	"t_q9_3"
##	[859]	"t_q9_4"	"t_q9_5"	"t_q10a_1"
##	[862]	"t_q10a_2"	"t_q10a_3"	"t_q10a_4"
##	[865]	"t_q10a_5"	"t_q10a_6"	"t_q10b_1"
##	[868]	"t_q10b_2"	"t_q10b_3"	"t_q10b_4"
##	[871]	"t_q10b_5"	"t_q10b_6"	"t_q10c_1"
##	[874]	"t_q10c_2"	"t_q10c_3"	"t_q10c_4"
##	[877]	"t_q10c_5"	"t_q10c_6"	"t_q10d_1"
##	[880]	"t_q10d_2"	"t_q10d_3"	"t_q10d_4"
##	[883]	"t_q10d_5"	"t_q10d_6"	"t_q10e_1"
##	[886]	"t_q10e_2"	"t_q10e_3"	"t_q10e_4"
##	[889]	"t_q10e_5"	"t_q10e_6"	"t_q11a_1"
##	[892]	"t_q11a_2"	"t_q11a_3"	"t_q11a_4"
##	[895]	"t_q11b_1"	"t_q11b_2"	"t_q11b_3"
##	[898]	"t_q11b_4"	"t_q11c_1"	"t_q11c_2"
##	[901]	"t_q11c_3"	"t_q11c_4"	"t_q11d_1"
##	[904]	"t_q11d_2"	"t_q11d_3"	"t_q11d_4"
##	[907]	"t_q11e_1"	"t_q11e_2"	"t_q11e_3"
##	[910]	"t_q11e_4"	"t_q11f_1"	"t_q11f_2"
##	[913]	"t_q11f_3"	"t_q11f_4"	"t_q11g_1"
##	[916]	"t_q11g_2"	"t_q11g_3"	"t_q11g_4"
##	[919]	"t_q11h_1"	"t_q11h_2"	"t_q11h_3"
##	[922]	"t_q11h_4"	"t_q11i_1"	"t_q11i_2"
##	[925]	"t_q11i_3"	"t_q11i_4"	"t_q11j_1"
##	[928]	"t_q11j_2"	"t_q11j_3"	"t_q11j_4"
##	[931]	"t_q11k_1"	"t_q11k_2"	"t_q11k_3"
##	[934]	"t_q11k_4"	"t_q11l_1"	"t_q11l_2"
##	[937]	"t_q11l_3"	"t_q11l_4"	"t_q11m_1"
##	[940]	"t_q11m_2"	"t_q11m_3"	"t_q11m_4"
##	[943]	"t_q11n_1"	"t_q11n_2"	"t_q11n_3"
##	[946]	"t_q11n_4"	"t_q11o_1"	"t_q11o_2"
##	[949]	"t_q11o_3"	"t_q11o_4"	"t_q12a_1"

##	[952]	"t_q12a_2"	"t_q12a_3"	"t_q12a_4"
##	[955]	"t_q12a_5"	"t_q12b_1"	"t_q12b_2"
##	[958]	"t_q12b_3"	"t_q12b_4"	"t_q12b_5"
##	[961]	"t_q12c_1"	"t_q12c_2"	"t_q12c_3"
##	[964]	"t_q12c_4"	"t_q12c_5"	"t_q12d_1"
##	[967]	"t_q12d_2"	"t_q12d_3"	"t_q12d_4"
##	[970]	"t_q12d_5"	"t_q13a_1"	"t_q13a_2"
##	[973]	"t_q13a_3"	"t_q13a_4"	"t_q13a_5"
##	[976]	"t_q13b_1"	"t_q13b_2"	"t_q13b_3"
##	[979]	"t_q13b_4"	"t_q13b_5"	"t_q13c_1"
##	[982]	"t_q13c_2"	"t_q13c_3"	"t_q13c_4"
##	[985]	"t_q13c_5"	"t_q13d_1"	"t_q13d_2"
##	[988]	"t_q13d_3"	"t_q13d_4"	"t_q13d_5"
##	[991]	"t_q14_1"	"t_q14_2"	"t_q14_3"
##	[994]	"t_q14_4"	"t_q14_5"	"t_q15a_1"
##	[997]	"t_q15a_2"	"t_q15a_3"	"t_q15a_4"
##	[1000]	"t_q15b_1"	"t_q15b_2"	"t_q15b_3"
##	[1003]	"t_q15b_4"	"t_q15c_1"	"t_q15c_2"
##	[1006]	"t_q15c_3"	"t_q15c_4"	"t_q15d_1"
##	[1009]	"t_q15d_2"	"t_q15d_3"	"t_q15d_4"
##	[1012]	"t_q15e_1"	"t_q15e_2"	"t_q15e_3"
##	[1015]	"t_q15e_4"	"t_q15f_1"	"t_q15f_2"
##	[1018]	"t_q15f_3"	"t_q15f_4"	"t_N_q1a_1"
##	[1021]	"t_N_q1a_2"	"t_N_q1a_3"	"t_N_q1a_4"
##	[1024]	"t_N_q1b_1"	"t_N_q1b_2"	"t_N_q1b_3"
##	[1027]	"t_N_q1b_4"	"t_N_q1c_1"	"t_N_q1c_2"
##	[1030]	"t_N_q1c_3"	"t_N_q1c_4"	"t_N_q1d_1"
##	[1033]	"t_N_q1d_2"	"t_N_q1d_3"	"t_N_q1d_4"
##	[1036]	"t_N_q1e_1"	"t_N_q1e_2"	"t_N_q1e_3"
##	[1039]	"t_N_q1e_4"	"t_N_q1f_1"	"t_N_q1f_2"
##	[1042]	"t_N_q1f_3"	"t_N_q1f_4"	"t_N_q1g_1"
##	[1045]	"t_N_q1g_2"	"t_N_q1g_3"	"t_N_q1g_4"
##	[1048]	"t_N_q2a_1"	"t_N_q2a_2"	"t_N_q2a_3"
##	[1051]	"t_N_q2a_4"	"t_N_q2b_1"	"t_N_q2b_2"
##	[1054]	"t_N_q2b_3"	"t_N_q2b_4"	"t_N_q2c_1"
##	[1057]	"t_N_q2c_2"	"t_N_q2c_3"	"t_N_q2c_4"
##	[1060]	"t_N_q2d_1"	"t_N_q2d_2"	"t_N_q2d_3"
##	[1063]	"t_N_q2d_4"	"t_N_q2e_1"	"t_N_q2e_2"
##	[1066]	"t_N_q2e_3"	"t_N_q2e_4"	"t_N_q2f_1"
##	[1069]	"t_N_q2f_2"	"t_N_q2f_3"	"t_N_q2f_4"
##	[1072]	"t_N_q3a_1"	"t_N_q3a_2"	"t_N_q3a_3"
##	[1075]	"t_N_q3b_1"	"t_N_q3b_2"	"t_N_q3b_3"
##	[1078]	"t_N_q3c_1"	"t_N_q3c_2"	"t_N_q3c_3"
##	[1081]	"t_N_q3d_1"	"t_N_q3d_2"	"t_N_q3d_3"
##	[1084]	"t_N_q3e_1"	"t_N_q3e_2"	"t_N_q3e_3"
##	[1087]	"t_N_q3f_1"	"t_N_q3f_2"	"t_N_q3f_3"
##	[1090]	"t_N_q3g_1"	"t_N_q3g_2"	"t_N_q3g_3"
##	[1093]	"t_N_q3h_1"	"t_N_q3h_2"	"t_N_q3h_3"
##	[1096]	"t_N_q3i_1"	"t_N_q3i_2"	"t_N_q3i_3"
##	[1099]	"t_N_q3j_1"	"t_N_q3j_2"	"t_N_q3j_3"
##	[1102]	"t_N_q4_1"	"t_N_q4_2"	"t_N_q4_3"
##	[1105]	"t_N_q4_4"	"t_N_q5a_1"	"t_N_q5a_2"
##	[1108]	"t_N_q5a_3"	"t_N_q5a_4"	"t_N_q5a_5"
##	[1111]	"t_N_q5b_1"	"t_N_q5b_2"	"t_N_q5b_3"

## [1114]	"t_N_q5b_4"	"t_N_q5b_5"	"t_N_q5c_1"
## [1117]	"t_N_q5c_2"	"t_N_q5c_3"	"t_N_q5c_4"
## [1120]	"t_N_q5c_5"	"t_N_q6a_1"	"t_N_q6a_2"
## [1123]	"t_N_q6a_3"	"t_N_q6a_4"	"t_N_q6b_1"
## [1126]	"t_N_q6b_2"	"t_N_q6b_3"	"t_N_q6b_4"
## [1129]	"t_N_q6c_1"	"t_N_q6c_2"	"t_N_q6c_3"
## [1132]	"t_N_q6c_4"	"t_N_q6d_1"	"t_N_q6d_2"
## [1135]	"t_N_q6d_3"	"t_N_q6d_4"	"t_N_q6e_1"
## [1138]	"t_N_q6e_2"	"t_N_q6e_3"	"t_N_q6e_4"
## [1141]	"t_N_q6f_1"	"t_N_q6f_2"	"t_N_q6f_3"
## [1144]	"t_N_q6f_4"	"t_N_q6g_1"	"t_N_q6g_2"
## [1147]	"t_N_q6g_3"	"t_N_q6g_4"	"t_N_q6h_1"
## [1150]	"t_N_q6h_2"	"t_N_q6h_3"	"t_N_q6h_4"
## [1153]	"t_N_q6i_1"	"t_N_q6i_2"	"t_N_q6i_3"
## [1156]	"t_N_q6i_4"	"t_N_q6j_1"	"t_N_q6j_2"
## [1159]	"t_N_q6j_3"	"t_N_q6j_4"	"t_N_q6k_1"
## [1162]	"t_N_q6k_2"	"t_N_q6k_3"	"t_N_q6k_4"
## [1165]	"t_N_q6l_1"	"t_N_q6l_2"	"t_N_q6l_3"
## [1168]	"t_N_q6l_4"	"t_N_q6m_1"	"t_N_q6m_2"
## [1171]	"t_N_q6m_3"	"t_N_q6m_4"	"t_N_q7a_1"
## [1174]	"t_N_q7a_2"	"t_N_q7a_3"	"t_N_q7a_4"
## [1177]	"t_N_q7a_5"	"t_N_q7b_1"	"t_N_q7b_2"
## [1180]	"t_N_q7b_3"	"t_N_q7b_4"	"t_N_q7b_5"
## [1183]	"t_N_q7c_1"	"t_N_q7c_2"	"t_N_q7c_3"
## [1186]	"t_N_q7c_4"	"t_N_q7c_5"	"t_N_q7d_1"
## [1189]	"t_N_q7d_2"	"t_N_q7d_3"	"t_N_q7d_4"
## [1192]	"t_N_q7d_5"	"t_N_q7e_1"	"t_N_q7e_2"
## [1195]	"t_N_q7e_3"	"t_N_q7e_4"	"t_N_q7e_5"
## [1198]	"t_N_q8a_1"	"t_N_q8a_2"	"t_N_q8a_3"
## [1201]	"t_N_q8a_4"	"t_N_q8b_1"	"t_N_q8b_2"
## [1204]	"t_N_q8b_3"	"t_N_q8b_4"	"t_N_q8c_1"
## [1207]	"t_N_q8c_2"	"t_N_q8c_3"	"t_N_q8c_4"
## [1210]	"t_N_q9_1"	"t_N_q9_2"	"t_N_q9_3"
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## [1216]	"t_N_q10a_2"	"t_N_q10a_3"	"t_N_q10a_4"
## [1219]	"t_N_q10a_5"	"t_N_q10a_6"	"t_N_q10b_1"
## [1222]	"t_N_q10b_2"	"t_N_q10b_3"	"t_N_q10b_4"
## [1225]	"t_N_q10b_5"	"t_N_q10b_6"	"t_N_q10c_1"
## [1228]	"t_N_q10c_2"	"t_N_q10c_3"	"t_N_q10c_4"
## [1231]	"t_N_q10c_5"	"t_N_q10c_6"	"t_N_q10d_1"
## [1234]	"t_N_q10d_2"	"t_N_q10d_3"	"t_N_q10d_4"
## [1237]	"t_N_q10d_5"	"t_N_q10d_6"	"t_N_q10e_1"
## [1240]	"t_N_q10e_2"	"t_N_q10e_3"	"t_N_q10e_4"
## [1243]	"t_N_q10e_5"	"t_N_q10e_6"	"t_N_q11a_1"
## [1246]	"t_N_q11a_2"	"t_N_q11a_3"	"t_N_q11a_4"
## [1249]	"t_N_q11b_1"	"t_N_q11b_2"	"t_N_q11b_3"
## [1252]	"t_N_q11b_4"	"t_N_q11c_1"	"t_N_q11c_2"
## [1255]	"t_N_q11c_3"	"t_N_q11c_4"	"t_N_q11d_1"
## [1258]	"t_N_q11d_2"	"t_N_q11d_3"	"t_N_q11d_4"
## [1261]	"t_N_q11e_1"	"t_N_q11e_2"	"t_N_q11e_3"
## [1264]	"t_N_q11e_4"	"t_N_q11f_1"	"t_N_q11f_2"
## [1267]	"t_N_q11f_3"	"t_N_q11f_4"	"t_N_q11g_1"
## [1270]	"t_N_q11g_2"	"t_N_q11g_3"	"t_N_q11g_4"
## [1273]	"t_N_q11h_1"	"t_N_q11h_2"	"t_N_q11h_3"

## [1276]	"t_N_q11h_4"	"t_N_q11i_1"	"t_N_q11i_2"
## [1279]	"t_N_q11i_3"	"t_N_q11i_4"	"t_N_q11j_1"
## [1282]	"t_N_q11j_2"	"t_N_q11j_3"	"t_N_q11j_4"
## [1285]	"t_N_q11k_1"	"t_N_q11k_2"	"t_N_q11k_3"
## [1288]	"t_N_q11k_4"	"t_N_q11l_1"	"t_N_q11l_2"
## [1291]	"t_N_q11l_3"	"t_N_q11l_4"	"t_N_q11m_1"
## [1294]	"t_N_q11m_2"	"t_N_q11m_3"	"t_N_q11m_4"
## [1297]	"t_N_q11n_1"	"t_N_q11n_2"	"t_N_q11n_3"
## [1300]	"t_N_q11n_4"	"t_N_q11o_1"	"t_N_q11o_2"
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## [1315]	"t_N_q12c_1"	"t_N_q12c_2"	"t_N_q12c_3"
## [1318]	"t_N_q12c_4"	"t_N_q12c_5"	"t_N_q12d_1"
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## [1327]	"t_N_q13a_3"	"t_N_q13a_4"	"t_N_q13a_5"
## [1330]	"t_N_q13b_1"	"t_N_q13b_2"	"t_N_q13b_3"
## [1333]	"t_N_q13b_4"	"t_N_q13b_5"	"t_N_q13c_1"
## [1336]	"t_N_q13c_2"	"t_N_q13c_3"	"t_N_q13c_4"
## [1339]	"t_N_q13c_5"	"t_N_q13d_1"	"t_N_q13d_2"
## [1342]	"t_N_q13d_3"	"t_N_q13d_4"	"t_N_q13d_5"
## [1345]	"t_N_q14_1"	"t_N_q14_2"	"t_N_q14_3"
## [1348]	"t_N_q14_4"	"t_N_q14_5"	"t_N_q15a_1"
## [1351]	"t_N_q15a_2"	"t_N_q15a_3"	"t_N_q15a_4"
## [1354]	"t_N_q15b_1"	"t_N_q15b_2"	"t_N_q15b_3"
## [1357]	"t_N_q15b_4"	"t_N_q15c_1"	"t_N_q15c_2"
## [1360]	"t_N_q15c_3"	"t_N_q15c_4"	"t_N_q15d_1"
## [1363]	"t_N_q15d_2"	"t_N_q15d_3"	"t_N_q15d_4"
## [1366]	"t_N_q15e_1"	"t_N_q15e_2"	"t_N_q15e_3"
## [1369]	"t_N_q15e_4"	"t_N_q15f_1"	"t_N_q15f_2"
## [1372]	"t_N_q15f_3"	"t_N_q15f_4"	"s_q6a"
## [1375]	"s_q6b"	"s_q6c"	"s_q12a"
## [1378]	"s_q12b"	"s_q12c"	"s_q13a"
## [1381]	"s_q13b"	"s_q13c"	"s_q13d"
## [1384]	"s_q13e"	"s_q13f"	"s_q13g"
## [1387]	"s_q14a"	"s_q14b"	"s_q14c"
## [1390]	"s_q14d"	"s_q14e"	"s_q14f"
## [1393]	"s_q14g"	"s_q2b"	"s_q4a"
## [1396]	"s_q4b"	"s_q8b"	"s_q8c"
## [1399]	"s_q8d"	"s_q2a"	"s_q2c"
## [1402]	"s_q5a"	"s_q5b"	"s_q6d"
## [1405]	"s_q6e"	"s_q6f"	"s_q6g"
## [1408]	"s_q6h"	"s_q9"	"s_q10"
## [1411]	"s_q11"	"s_q3a"	"s_q3b"
## [1414]	"s_q3c"	"s_q3d"	"s_q3e"
## [1417]	"s_q3f"	"s_q3g"	"s_q3h"
## [1420]	"s_q3i"	"s_q7a"	"s_q7b"
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## [1426]	"s_q1_2"	"s_q1_3"	"s_q1_4"
## [1429]	"s_q1_5"	"s_q1_6"	"s_q1_7"
## [1432]	"s_q2a_1"	"s_q2a_2"	"s_q2a_3"
## [1435]	"s_q2a_4"	"s_q2b_1"	"s_q2b_2"

## [1438]	"s_q2b_3"	"s_q2b_4"	"s_q2c_1"
## [1441]	"s_q2c_2"	"s_q2c_3"	"s_q2c_4"
## [1444]	"s_q3a_1"	"s_q3a_2"	"s_q3a_3"
## [1447]	"s_q3a_4"	"s_q3b_1"	"s_q3b_2"
## [1450]	"s_q3b_3"	"s_q3b_4"	"s_q3c_1"
## [1453]	"s_q3c_2"	"s_q3c_3"	"s_q3c_4"
## [1456]	"s_q3d_1"	"s_q3d_2"	"s_q3d_3"
## [1459]	"s_q3d_4"	"s_q3e_1"	"s_q3e_2"
## [1462]	"s_q3e_3"	"s_q3e_4"	"s_q3f_1"
## [1465]	"s_q3f_2"	"s_q3f_3"	"s_q3f_4"
## [1468]	"s_q3g_1"	"s_q3g_2"	"s_q3g_3"
## [1471]	"s_q3g_4"	"s_q3h_1"	"s_q3h_2"
## [1474]	"s_q3h_3"	"s_q3h_4"	"s_q3i_1"
## [1477]	"s_q3i_2"	"s_q3i_3"	"s_q3i_4"
## [1480]	"s_q4a_1"	"s_q4a_2"	"s_q4a_3"
## [1483]	"s_q4a_4"	"s_q4b_1"	"s_q4b_2"
## [1486]	"s_q4b_3"	"s_q4b_4"	"s_q5a_1"
## [1489]	"s_q5a_2"	"s_q5a_3"	"s_q5a_4"
## [1492]	"s_q5b_1"	"s_q5b_2"	"s_q5b_3"
## [1495]	"s_q5b_4"	"s_q6a_1"	"s_q6a_2"
## [1498]	"s_q6a_3"	"s_q6a_4"	"s_q6a_5"
## [1501]	"s_q6b_1"	"s_q6b_2"	"s_q6b_3"
## [1504]	"s_q6b_4"	"s_q6b_5"	"s_q6c_1"
## [1507]	"s_q6c_2"	"s_q6c_3"	"s_q6c_4"
## [1510]	"s_q6c_5"	"s_q6d_1"	"s_q6d_2"
## [1513]	"s_q6d_3"	"s_q6d_4"	"s_q6d_5"
## [1516]	"s_q6e_1"	"s_q6e_2"	"s_q6e_3"
## [1519]	"s_q6e_4"	"s_q6e_5"	"s_q6f_1"
## [1522]	"s_q6f_2"	"s_q6f_3"	"s_q6f_4"
## [1525]	"s_q6f_5"	"s_q6g_1"	"s_q6g_2"
## [1528]	"s_q6g_3"	"s_q6g_4"	"s_q6g_5"
## [1531]	"s_q6h_1"	"s_q6h_2"	"s_q6h_3"
## [1534]	"s_q6h_4"	"s_q6h_5"	"s_q7a_1"
## [1537]	"s_q7a_2"	"s_q7a_3"	"s_q7a_4"
## [1540]	"s_q7b_1"	"s_q7b_2"	"s_q7b_3"
## [1543]	"s_q7b_4"	"s_q8a_1"	"s_q8a_2"
## [1546]	"s_q8a_3"	"s_q8a_4"	"s_q8a_5"
## [1549]	"s_q8b_1"	"s_q8b_2"	"s_q8b_3"
## [1552]	"s_q8b_4"	"s_q8b_5"	"s_q8c_1"
## [1555]	"s_q8c_2"	"s_q8c_3"	"s_q8c_4"
## [1558]	"s_q8c_5"	"s_q8d_1"	"s_q8d_2"
## [1561]	"s_q8d_3"	"s_q8d_4"	"s_q8d_5"
## [1564]	"s_q9a_1"	"s_q9a_2"	"s_q9a_3"
## [1567]	"s_q9b_1"	"s_q9b_2"	"s_q9b_3"
## [1570]	"s_q9c_1"	"s_q9c_2"	"s_q9c_3"
## [1573]	"s_q9d_1"	"s_q9d_2"	"s_q9d_3"
## [1576]	"s_q9e_1"	"s_q9e_2"	"s_q9e_3"
## [1579]	"s_q9f_1"	"s_q9f_2"	"s_q9f_3"
## [1582]	"s_q9g_1"	"s_q9g_2"	"s_q9g_3"
## [1585]	"s_q9h_1"	"s_q9h_2"	"s_q9h_3"
## [1588]	"s_q10a_1"	"s_q10a_2"	"s_q10a_3"
## [1591]	"s_q10b_1"	"s_q10b_2"	"s_q10b_3"
## [1594]	"s_q10c_1"	"s_q10c_2"	"s_q10c_3"
## [1597]	"s_q10d_1"	"s_q10d_2"	"s_q10d_3"

## [1600]	"s_q10e_1"	"s_q10e_2"	"s_q10e_3"
## [1603]	"s_q10f_1"	"s_q10f_2"	"s_q10f_3"
## [1606]	"s_q10g_1"	"s_q10g_2"	"s_q10g_3"
## [1609]	"s_q10h_1"	"s_q10h_2"	"s_q10h_3"
## [1612]	"s_q11_1"	"s_q11_2"	"s_q11_3"
## [1615]	"s_q11_4"	"s_q12a_1"	"s_q12a_2"
## [1618]	"s_q12a_3"	"s_q12a_4"	"s_q12b_1"
## [1621]	"s_q12b_2"	"s_q12b_3"	"s_q12b_4"
## [1624]	"s_q12c_1"	"s_q12c_2"	"s_q12c_3"
## [1627]	"s_q12c_4"	"s_q13a_1"	"s_q13a_2"
## [1630]	"s_q13a_3"	"s_q13a_4"	"s_q13b_1"
## [1633]	"s_q13b_2"	"s_q13b_3"	"s_q13b_4"
## [1636]	"s_q13c_1"	"s_q13c_2"	"s_q13c_3"
## [1639]	"s_q13c_4"	"s_q13d_1"	"s_q13d_2"
## [1642]	"s_q13d_3"	"s_q13d_4"	"s_q13e_1"
## [1645]	"s_q13e_2"	"s_q13e_3"	"s_q13e_4"
## [1648]	"s_q13f_1"	"s_q13f_2"	"s_q13f_3"
## [1651]	"s_q13f_4"	"s_q13g_1"	"s_q13g_2"
## [1654]	"s_q13g_3"	"s_q13g_4"	"s_q14a_1"
## [1657]	"s_q14a_2"	"s_q14a_3"	"s_q14a_4"
## [1660]	"s_q14b_1"	"s_q14b_2"	"s_q14b_3"
## [1663]	"s_q14b_4"	"s_q14c_1"	"s_q14c_2"
## [1666]	"s_q14c_3"	"s_q14c_4"	"s_q14d_1"
## [1669]	"s_q14d_2"	"s_q14d_3"	"s_q14d_4"
## [1672]	"s_q14e_1"	"s_q14e_2"	"s_q14e_3"
## [1675]	"s_q14e_4"	"s_q14f_1"	"s_q14f_2"
## [1678]	"s_q14f_3"	"s_q14f_4"	"s_q14g_1"
## [1681]	"s_q14g_2"	"s_q14g_3"	"s_q14g_4"
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## [1687]	"s_N_q1_4"	"s_N_q1_5"	"s_N_q1_6"
## [1690]	"s_N_q1_7"	"s_N_q2a_1"	"s_N_q2a_2"
## [1693]	"s_N_q2a_3"	"s_N_q2a_4"	"s_N_q2b_1"
## [1696]	"s_N_q2b_2"	"s_N_q2b_3"	"s_N_q2b_4"
## [1699]	"s_N_q2c_1"	"s_N_q2c_2"	"s_N_q2c_3"
## [1702]	"s_N_q2c_4"	"s_N_q3a_1"	"s_N_q3a_2"
## [1705]	"s_N_q3a_3"	"s_N_q3a_4"	"s_N_q3b_1"
## [1708]	"s_N_q3b_2"	"s_N_q3b_3"	"s_N_q3b_4"
## [1711]	"s_N_q3c_1"	"s_N_q3c_2"	"s_N_q3c_3"
## [1714]	"s_N_q3c_4"	"s_N_q3d_1"	"s_N_q3d_2"
## [1717]	"s_N_q3d_3"	"s_N_q3d_4"	"s_N_q3e_1"
## [1720]	"s_N_q3e_2"	"s_N_q3e_3"	"s_N_q3e_4"
## [1723]	"s_N_q3f_1"	"s_N_q3f_2"	"s_N_q3f_3"
## [1726]	"s_N_q3f_4"	"s_N_q3g_1"	"s_N_q3g_2"
## [1729]	"s_N_q3g_3"	"s_N_q3g_4"	"s_N_q3h_1"
## [1732]	"s_N_q3h_2"	"s_N_q3h_3"	"s_N_q3h_4"
## [1735]	"s_N_q3i_1"	"s_N_q3i_2"	"s_N_q3i_3"
## [1738]	"s_N_q3i_4"	"s_N_q4a_1"	"s_N_q4a_2"
## [1741]	"s_N_q4a_3"	"s_N_q4a_4"	"s_N_q4b_1"
## [1744]	"s_N_q4b_2"	"s_N_q4b_3"	"s_N_q4b_4"
## [1747]	"s_N_q5a_1"	"s_N_q5a_2"	"s_N_q5a_3"
## [1750]	"s_N_q5a_4"	"s_N_q5b_1"	"s_N_q5b_2"
## [1753]	"s_N_q5b_3"	"s_N_q5b_4"	"s_N_q6a_1"
## [1756]	"s_N_q6a_2"	"s_N_q6a_3"	"s_N_q6a_4"
## [1759]	"s_N_q6a_5"	"s_N_q6b_1"	"s_N_q6b_2"

## [1762]	"s_N_q6b_3"	"s_N_q6b_4"	"s_N_q6b_5"
## [1765]	"s_N_q6c_1"	"s_N_q6c_2"	"s_N_q6c_3"
## [1768]	"s_N_q6c_4"	"s_N_q6c_5"	"s_N_q6d_1"
## [1771]	"s_N_q6d_2"	"s_N_q6d_3"	"s_N_q6d_4"
## [1774]	"s_N_q6d_5"	"s_N_q6e_1"	"s_N_q6e_2"
## [1777]	"s_N_q6e_3"	"s_N_q6e_4"	"s_N_q6e_5"
## [1780]	"s_N_q6f_1"	"s_N_q6f_2"	"s_N_q6f_3"
## [1783]	"s_N_q6f_4"	"s_N_q6f_5"	"s_N_q6g_1"
## [1786]	"s_N_q6g_2"	"s_N_q6g_3"	"s_N_q6g_4"
## [1789]	"s_N_q6g_5"	"s_N_q6h_1"	"s_N_q6h_2"
## [1792]	"s_N_q6h_3"	"s_N_q6h_4"	"s_N_q6h_5"
## [1795]	"s_N_q7a_1"	"s_N_q7a_2"	"s_N_q7a_3"
## [1798]	"s_N_q7a_4"	"s_N_q7b_1"	"s_N_q7b_2"
## [1801]	"s_N_q7b_3"	"s_N_q7b_4"	"s_N_q8a_1"
## [1804]	"s_N_q8a_2"	"s_N_q8a_3"	"s_N_q8a_4"
## [1807]	"s_N_q8a_5"	"s_N_q8b_1"	"s_N_q8b_2"
## [1810]	"s_N_q8b_3"	"s_N_q8b_4"	"s_N_q8b_5"
## [1813]	"s_N_q8c_1"	"s_N_q8c_2"	"s_N_q8c_3"
## [1816]	"s_N_q8c_4"	"s_N_q8c_5"	"s_N_q8d_1"
## [1819]	"s_N_q8d_2"	"s_N_q8d_3"	"s_N_q8d_4"
## [1822]	"s_N_q8d_5"	"s_N_q9a_1"	"s_N_q9a_2"
## [1825]	"s_N_q9a_3"	"s_N_q9b_1"	"s_N_q9b_2"
## [1828]	"s_N_q9b_3"	"s_N_q9c_1"	"s_N_q9c_2"
## [1831]	"s_N_q9c_3"	"s_N_q9d_1"	"s_N_q9d_2"
## [1834]	"s_N_q9d_3"	"s_N_q9e_1"	"s_N_q9e_2"
## [1837]	"s_N_q9e_3"	"s_N_q9f_1"	"s_N_q9f_2"
## [1840]	"s_N_q9f_3"	"s_N_q9g_1"	"s_N_q9g_2"
## [1843]	"s_N_q9g_3"	"s_N_q9h_1"	"s_N_q9h_2"
## [1846]	"s_N_q9h_3"	"s_N_q10a_1"	"s_N_q10a_2"
## [1849]	"s_N_q10a_3"	"s_N_q10b_1"	"s_N_q10b_2"
## [1852]	"s_N_q10b_3"	"s_N_q10c_1"	"s_N_q10c_2"
## [1855]	"s_N_q10c_3"	"s_N_q10d_1"	"s_N_q10d_2"
## [1858]	"s_N_q10d_3"	"s_N_q10e_1"	"s_N_q10e_2"
## [1861]	"s_N_q10e_3"	"s_N_q10f_1"	"s_N_q10f_2"
## [1864]	"s_N_q10f_3"	"s_N_q10g_1"	"s_N_q10g_2"
## [1867]	"s_N_q10g_3"	"s_N_q10h_1"	"s_N_q10h_2"
## [1870]	"s_N_q10h_3"	"s_N_q11_1"	"s_N_q11_2"
## [1873]	"s_N_q11_3"	"s_N_q11_4"	"s_N_q12a_1"
## [1876]	"s_N_q12a_2"	"s_N_q12a_3"	"s_N_q12a_4"
## [1879]	"s_N_q12b_1"	"s_N_q12b_2"	"s_N_q12b_3"
## [1882]	"s_N_q12b_4"	"s_N_q12c_1"	"s_N_q12c_2"
## [1885]	"s_N_q12c_3"	"s_N_q12c_4"	"s_N_q13a_1"
## [1888]	"s_N_q13a_2"	"s_N_q13a_3"	"s_N_q13a_4"
## [1891]	"s_N_q13b_1"	"s_N_q13b_2"	"s_N_q13b_3"
## [1894]	"s_N_q13b_4"	"s_N_q13c_1"	"s_N_q13c_2"
## [1897]	"s_N_q13c_3"	"s_N_q13c_4"	"s_N_q13d_1"
## [1900]	"s_N_q13d_2"	"s_N_q13d_3"	"s_N_q13d_4"
## [1903]	"s_N_q13e_1"	"s_N_q13e_2"	"s_N_q13e_3"
## [1906]	"s_N_q13e_4"	"s_N_q13f_1"	"s_N_q13f_2"
## [1909]	"s_N_q13f_3"	"s_N_q13f_4"	"s_N_q13g_1"
## [1912]	"s_N_q13g_2"	"s_N_q13g_3"	"s_N_q13g_4"
## [1915]	"s_N_q14a_1"	"s_N_q14a_2"	"s_N_q14a_3"
## [1918]	"s_N_q14a_4"	"s_N_q14b_1"	"s_N_q14b_2"
## [1921]	"s_N_q14b_3"	"s_N_q14b_4"	"s_N_q14c_1"


```
## [1924] "s_N_q14c_2"      "s_N_q14c_3"      "s_N_q14c_4"
## [1927] "s_N_q14d_1"      "s_N_q14d_2"      "s_N_q14d_3"
## [1930] "s_N_q14d_4"      "s_N_q14e_1"      "s_N_q14e_2"
## [1933] "s_N_q14e_3"      "s_N_q14e_4"      "s_N_q14f_1"
## [1936] "s_N_q14f_2"      "s_N_q14f_3"      "s_N_q14f_4"
## [1939] "s_N_q14g_1"      "s_N_q14g_2"      "s_N_q14g_3"
## [1942] "s_N_q14g_4"
```

Simplifying the Dataframes

```
# select only relevant variables from df_combined
```

```
agg_cols <- c("saf_p_11",
  "com_p_11",
  "eng_p_11",
  "aca_p_11",
  "saf_t_11",
  "com_t_11",
  "eng_t_11",
  "aca_t_11",
  "saf_s_11",
  "com_s_11",
  "eng_s_11",
  "aca_s_11",
  "saf_tot_11",
  "com_tot_11",
  "eng_tot_11",
  "aca_tot_11")

df_d75 <- df_d75 %>%
  select("dbn", all_of(agg_cols), schooltype)

df_gened <- df_gened %>%
  select("dbn", all_of(agg_cols), schooltype)
```

Simplifying the Dataframes

- cbind is used to stack rows with the same column names on top of each other.
- left joining the survey data onto the combined data because we are asking whether survey perceptions *relate to* academic outcomes. All the rows in combined are retained, columns are added from the other datasets and where there is no corresponding value, NA is used.

```
# select only relevant variables from df_combined
# Step 1: simplify survey dataframes - done once, no duplication
agg_cols <- c("saf_p_11", "com_p_11", "eng_p_11", "aca_p_11",
  "saf_t_11", "com_t_11", "eng_t_11", "aca_t_11",
  "saf_s_11", "com_s_11", "eng_s_11", "aca_s_11",
  "saf_tot_11", "com_tot_11", "eng_tot_11", "aca_tot_11")

df_d75 <- df_d75 %>%
```

```

    select(dbn, schooltype, all_of(agg_cols))

df_gened <- df_gened %>%
  select(dbn, schooltype, all_of(agg_cols))

# Step 2: combine survey dataframes
df_survey <- bind_rows(df_gened, df_d75)

# Step 3: rename dbn to match combined, then left_join
df_survey <- df_survey %>%
  rename(DBN = dbn)

df_main <- df_combined %>%
  left_join(df_survey, by = "DBN")

# Step 4: pivot to long format
df_long <- df_main %>%
  pivot_longer(cols = all_of(agg_cols),
               names_to = "metric",
               values_to = "score")

# clean up the column names with janitor
df_long <- df_long %>% clean_names()

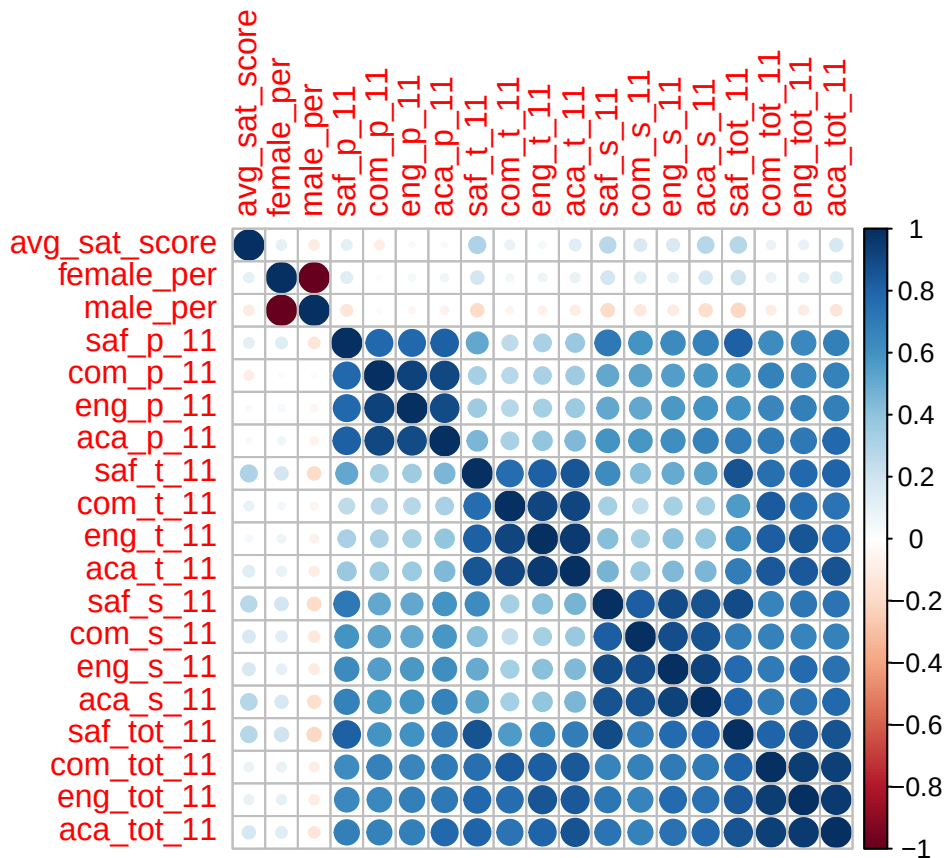
```

Correlations between variables

```

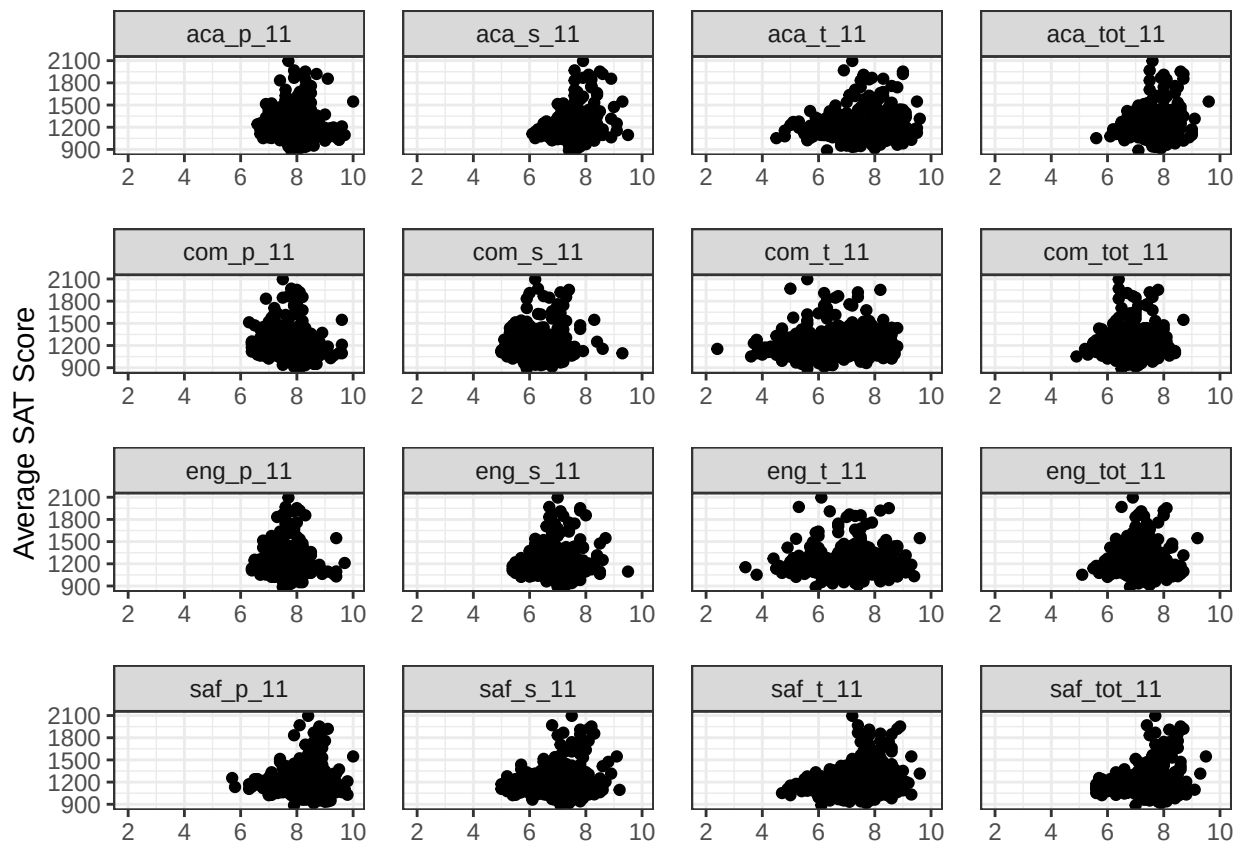
df_main %>%
  select(avg_sat_score, female_per, male_per,
         all_of(agg_cols)) %>%
  cor(use = "pairwise.complete.obs") %>%
  corrplot()

```



```
# tibble of correlations between variables, orded by absolute value descending
cor_tibble <- df_main %>%
  select(avg_sat_score, all_of(agg_cols)) %>%
  cor(use = "pairwise.complete.obs") %>%
  as_tibble(rownames = "variable") %>%
  pivot_longer(-variable, names_to = "variable2", values_to = "correlation") %>%
  filter(variable == "avg_sat_score") %>% # focus on SAT correlations
  arrange(desc(abs(correlation)))

df_main %>%
  select(avg_sat_score, all_of(agg_cols)) %>%
  drop_na() %>%
  pivot_longer(-avg_sat_score, names_to = "metric", values_to = "score") %>%
  ggplot(aes(x = score, y = avg_sat_score)) +
  geom_point() +
  facet_wrap(metric ~ ., scales = "free_x") +
  labs(y = "Average SAT Score", x = NULL) +
  coord_cartesian(xlim = c(1.9, 10)) +
  theme_bw() +
  theme(panel.spacing = unit(1, "lines"))
```



Results

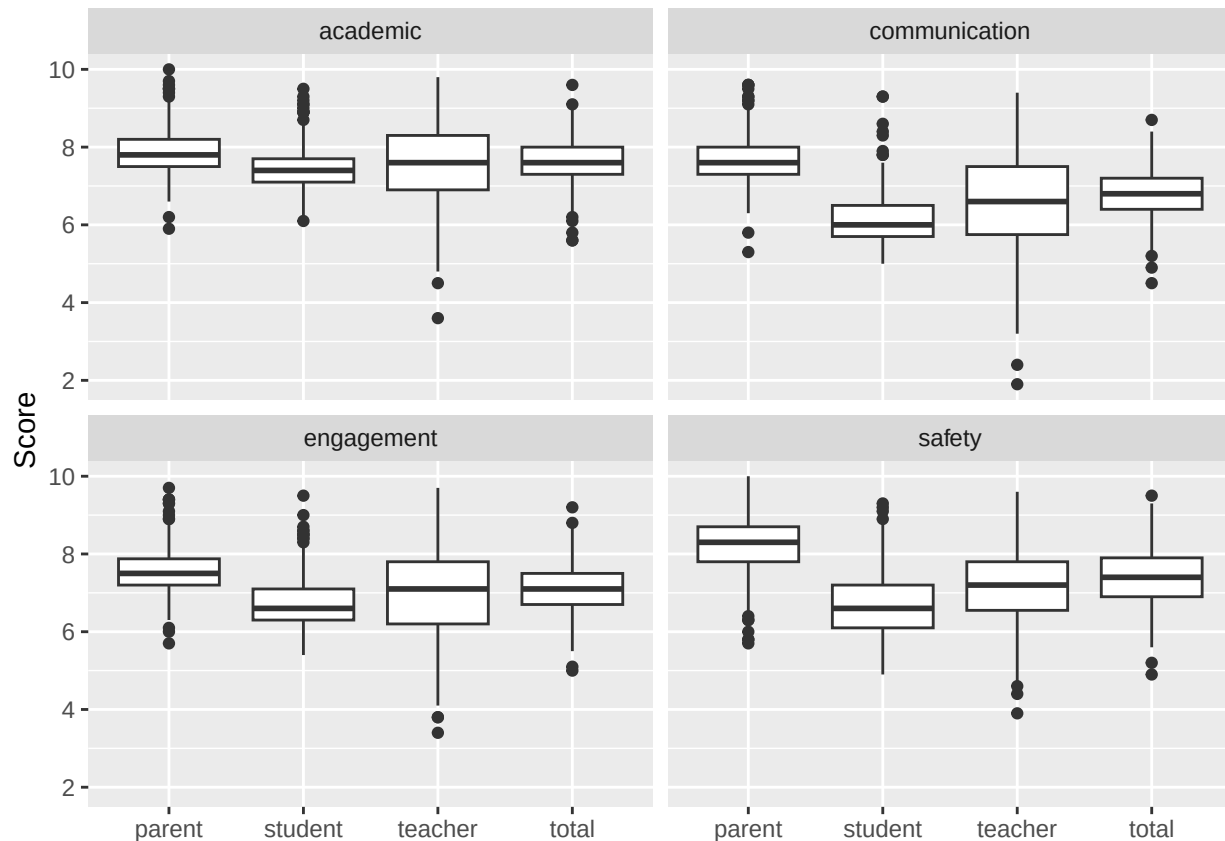
- scores range from 1.9 to 10.
- within the parents groups and student groups, they tend to rate similarly around 5-9 whereas teachers have a more spread out range of scores from 4 to 10 across all aggregated groups.

```
df_long <- df_long %>%
  mutate(
    survey_question = case_when(
      str_detect(metric, "saf") ~ "safety",
      str_detect(metric, "com") ~ "communication",
      str_detect(metric, "eng") ~ "engagement",
      str_detect(metric, "aca") ~ "academic"
    ),
    respondent_type = case_when(
      str_detect(metric, "_p_") ~ "parent",
      str_detect(metric, "_t_") ~ "teacher",
      str_detect(metric, "_s_") ~ "student",
      str_detect(metric, "_tot") ~ "total"
    )
  ) %>%
  select(dbn, school_name, survey_question, respondent_type, schooltype, metric, score)

ggplot(df_long, aes(x = respondent_type, y = score)) +
  geom_boxplot() +
  facet_wrap(~ survey_question) +
```

```
labs(x = NULL, y = "Score")
```

```
## Warning: Removed 148 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Findings:

- parents score highest in all question types and significantly higher scores from the other groups in communication, safety and engagement (the IQR does not overlap the other groups')
- students score lowest, especially in communication and safety
- teachers score between parents and students
- the most consistent scores are in academic, suggesting consistent perceptions of academic quality.
- the largest gap between parents and students is in communication. Teachers show a wide IQR, suggesting an internally wide range of perceptions.
- in engagement, parents are significantly higher than students, even though students are the ones who attend. The teacher IQR is wide again (see previous point)
- parents rate safety as significantly higher than the teachers and students, even though the students are the ones affected by it directly.

Conclusions

Parents consistently have a higher perception of the schools than teachers and students. Differences between groups are largest in communication and safety. This suggests that parents are not aware of the daily realities, or students and teachers have higher expectations than parents.